

Jason Zhi Liang, Ph.D.

AI Research Scientist | Principal Architect | Evolutionary Computation & Agentic Reasoning

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Summary

AI Research Scientist and Principal Architect working at the intersection of **evolutionary optimization**, **inference-time compute**, and **agentic reasoning**. Inventor of **CoDeepNEAT** (evolutionary neural architecture search) and architect of **Caesar**, an autonomous deep-research agent that scales test-time compute via adversarial self-refinement over a self-organizing knowledge graph.

Core Expertise

Research: Inference-Time Compute Scaling • Agentic Reasoning (System 2) • Evolutionary Population-Based Optimization • Neural Architecture Search • Neuroevolution • Meta-Learning • Recursive Self-Improvement

Agent Infrastructure: Graph-Augmented RAG • Stateful Agentic Memory • Multi-Agent Orchestration • FSM-Based Agent Coordination • LLM-as-a-Judge Evaluation • Evolutionary Prompt Optimization

Tooling: Python • Keras/TensorFlow • Frontier LLM APIs • AutoGen / MetaGPT • Distributed GPU Clusters

Professional Experience

Cognizant AI Labs

San Francisco, CA

Research Scientist & Principal Architect

Dec 2018 – Present

- Architected **Caesar**, an autonomous research agent (analog of Gemini Deep Research) that scales inference-time compute via an adversarial refinement loop—critiquing drafts, issuing orthogonal queries, and consolidating findings through generative merging.
- Replaced static RAG with a dynamic “Perceive-Think-Act” loop backed by a self-organizing knowledge graph, enabling associative reasoning across cross-disciplinary connections and overcoming the “navigational amnesia” of linear agents.
- Built a **Finite State Machine framework** coordinating specialized LLM agents; applied evolutionary search over team composition to achieve strong results on SciCode and HumanEval+.
- Pioneered **Evolutionary Population-Based Training (EPBT)**, evolving hyperparameters and loss functions during training to cut tuning compute vs. grid search on vision and language models.
- Created **Code Archaeologist**, an agentic system reasoning over Git/LFS repository histories to surface architectural patterns and technical debt.

Sentient Technologies

San Francisco, CA

Research Scientist (prev. Research Intern)

Dec 2015 – Nov 2018

- Invented and scaled **CoDeepNEAT** (evolutionary neural architecture search) to clusters of hundreds of GPUs, validating that evolutionary search scales with parallel compute—a key property for pre-training large models.
- Achieved state-of-the-art image captioning and object recognition by evolving neural topologies that share weights across tasks; designed and shipped production AutoML systems.

The University of Texas at Austin

Austin, TX

Ph.D. Candidate / Research Assistant

Aug 2013 – Dec 2018

- Developed **MEA**, a meta-evolutionary bilevel optimization framework that solved sparse-reward continuous control tasks (e.g., helicopter hovering) where gradient-based RL failed to converge; created the original **CoDeepNEAT** algorithm.

Education

Ph.D., Computer Science — University of Texas at Austin

2013 – 2018

Neuroevolution & Deep Learning. Advisor: Risto Miikkulainen. Dissertation: *Evolutionary Neural Architecture Search for Deep Learning*.

B.S., Electrical Engineering & Computer Science — University of California, Berkeley

2009 – 2013

Selected Publications & Patents

- Caesar: Deep Agentic Web Exploration for Creative Answer Synthesis** (2026) — Liang, Meyerson, Miikkulainen. *Under review, ICML 2026*.
- Asynchronous Evolution of Deep Neural Network Architectures** (2024) — Liang, Shahrzad, Miikkulainen. *Applied Soft Computing, Vol. 152*.
- Regularized Evolutionary Population-Based Training** (2021) — Liang et al. *GECCO 2021*.
- Evolutionary Architecture Search for Deep Multitask Networks** (2018) — Liang, Meyerson, Miikkulainen. *GECCO 2018* (introduced CoDeepNEAT).
- Patents:** US 11,507,844 (2022) • US 11,030,529 (2021) • US 11,003,994 (2021) • 5+ additional patent applications on NN evolution.

Awards & Service

BEACON Research Grant (2015), NSF Center for the Study of Evolution in Action. • **Reviewer**, GECCO (2019 – Present).